Hardware Security of DNNs: Fault-injection attacks

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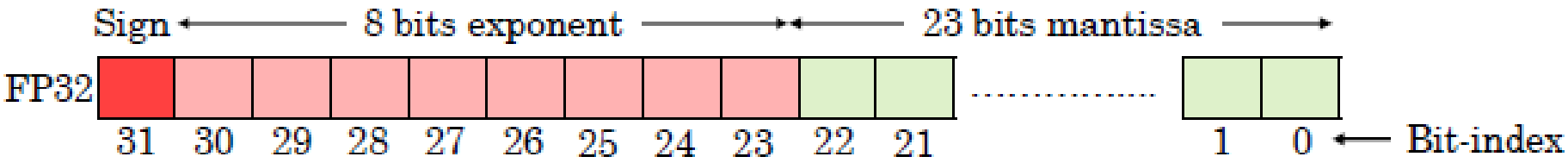
Vulnerability of different bits/layers/parameters

- In floating-point weights, attacking mantissa bits has little impact. Attacking exponent bits has a huge impact



- The attack on initial layers has higher impact than that on later layers
  - Errors in early layers accumulate and amplify
- In DNNs with ReLU, only positive parameters are vulnerable

Impact of 1→0 vs 0→1 bit-flip

- Only 0→ 1 flip causes large accuracy loss.
- Due to normal distribution of parameters, most parameters are inside the range [-1,1].
- ➔ 1→ 0 flip in the exponent bit reduces a parameter's magnitude by no more than one.
- 0→ 1 or 1→ 0 flips in sign bit alter the magnitude by no more than two.
- However, a 0→1 flip in exponent bits dramatically increases parameter value.
- ➔ During inference, unduly high activation produced by the faulty parameter overrides the remaining activations

BFA attack

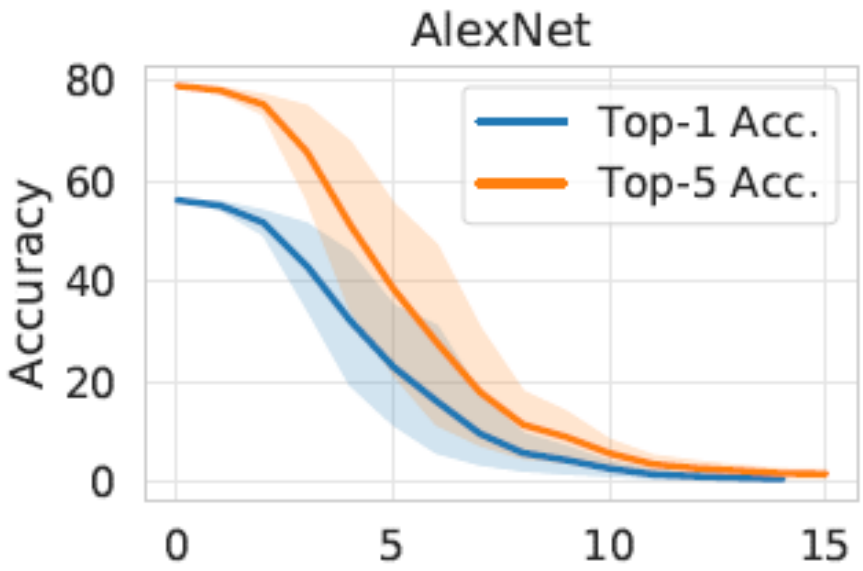
References:

- “Bit-Flip Attack: Crushing Neural Network with Progressive Bit Search”, ICCV 2019
- He et al, "Defending and Harnessing the Bit-Flip based Adversarial Weight Attack", CVPR, 2020
- Li et al., "RADAR: Run-time Adversarial Weight Attack Detection and Accuracy Recovery", Arxiv, 2021
- Li et al., "Defending Bit-Flip Attack through DNN Weight Reconstruction", DAC, 2020

Bit-flip attack (BFA)

- They quantize the weights to 8-bit weights and show them in 2's complement representation
- Their technique finds vulnerable bits in DNN weights to be flipped for highest loss in accuracy.
- These bits occur in weights with high gradient
- It takes low-magnitude weights and dramatically increases their value. It flips their MSBs.

Results of BFA



Model	Capacity (bits)	Baseline top-1 acc.	Bit-flips required
AlexNet	488M	56.1%	17
ResNet-50	204M	75.8%	11

Understanding characteristics of BFA attack

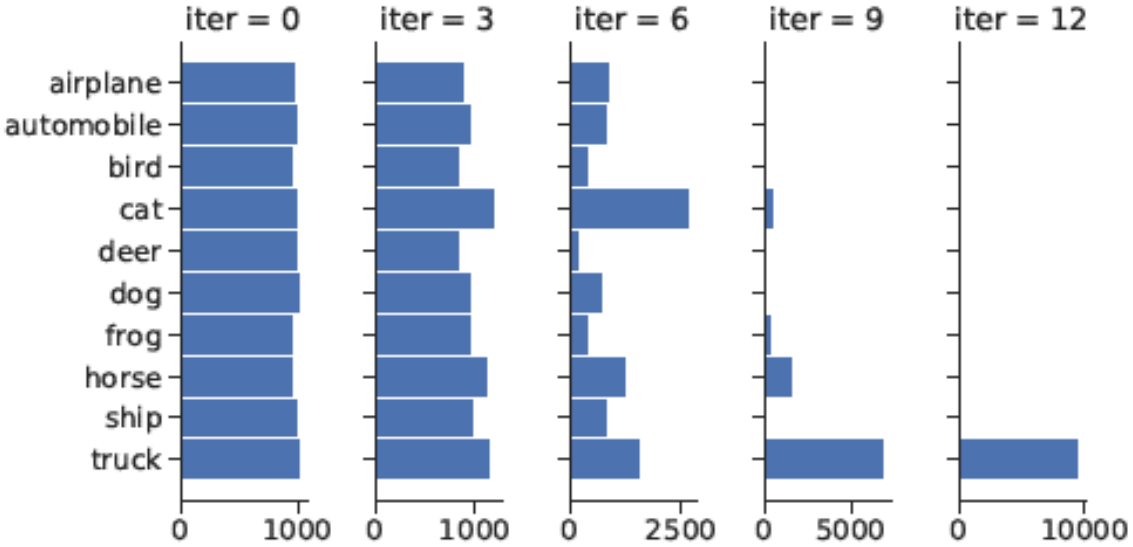
Some characteristics

- It tends to attack a small weight value (near-zero) and makes it a large value
- It attacks weights in front-end layers
- It attacks MSBs
  - In a sample CNN, attacking 10 MSBs degrades accuracy by same amount as attacking 30 MSB-1 bits

- **Defense:** Since only MSB is attacked, we can backup all the MSBs
  - Limitation: overhead of 12.5% (each weight is 8b)

Output category after attack

- In zeroth iteration of attack, classification predictions are uniformly spread in a different categories.
- With increasing # of iterations, nearly all inputs are predicted to belong to one category. This category depends on the attack sample data



ResNet-20 on CIFAR-10

Defense techniques

Defense techniques

- Piecewise clustering
- Binarization
- Increasing network capacity
- Batch-normalization and dropout
- Checksum+interleaving
- Averaging+quantization+clipping

Averaging+quantization+clipping defense technique

Observation: BFA causes large change in magnitude of a weight.
Key Idea of this technique: Reduce the change in weight value on an attack

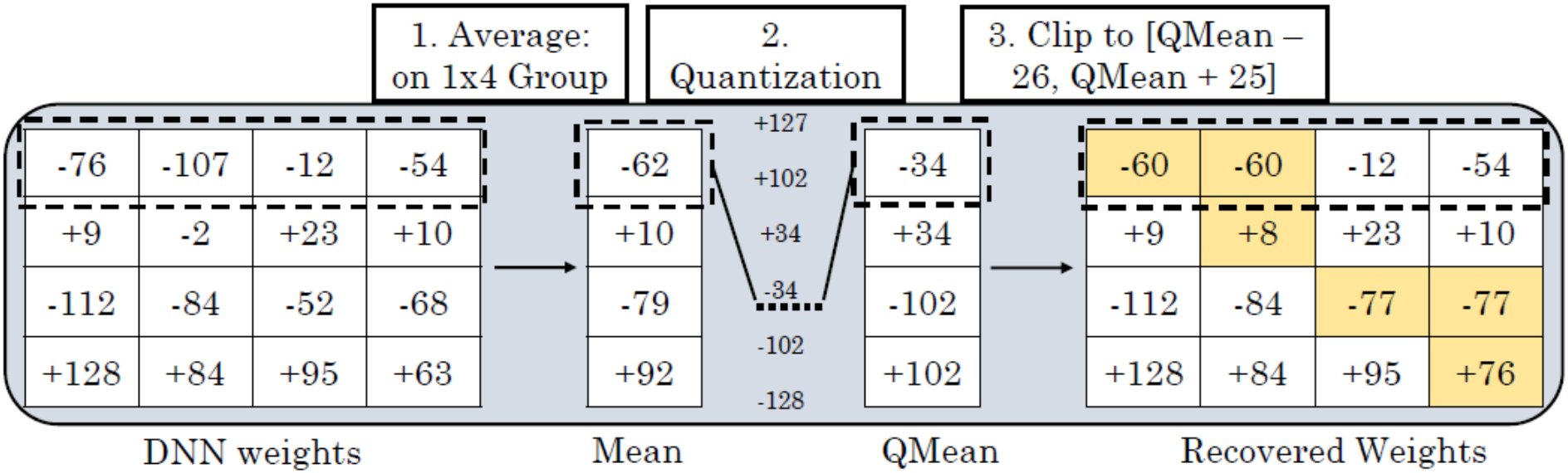
Averaging and quantization

- Assume, BFA changes a weight by Δ
- **Averaging**: Use a group of size G. On computing mean, the impact of BFA on mean is only Δ/G
- **Quantization:** seeks to cancel effect of Δ/G on the quantized mean (Qmean).
- With 8-bit weight, average can lie between -128 to 127.
- Hence, if the gap between two quantization levels is very large, quantization is highly likely to cancel Δ/G

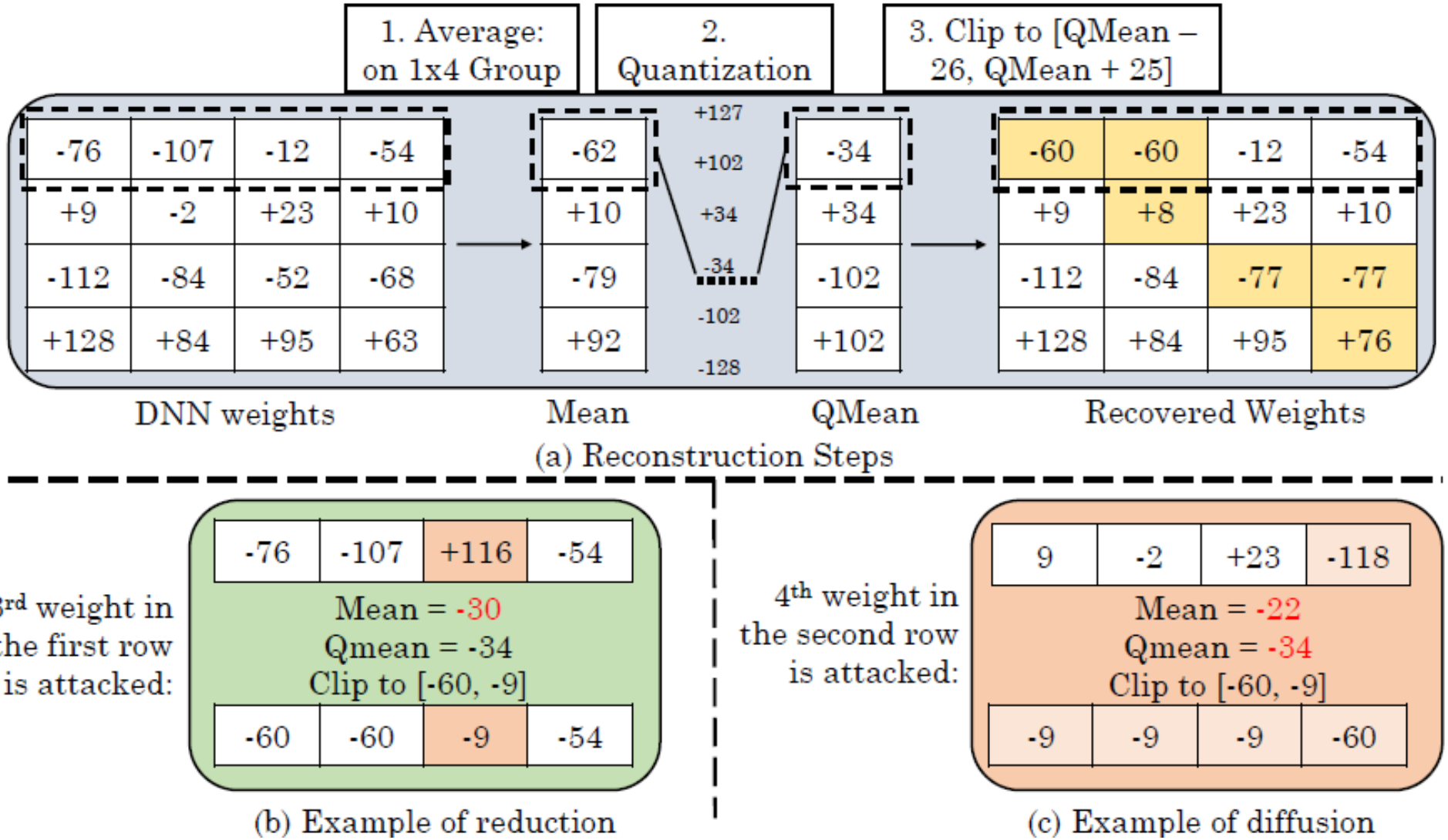
Clipping

- Clipping: it restricts all the weights to a small range around Qmean, say Φ
- In other words, maximum deviation of target weight from the cleaned model is within range, (Qmean-Φ, Qmean+Φ)

Example (1 of 2)



Example (2 of 2)



Averaging+quantization+clipping: ResNet18 Results

- The baseline model accuracy: 69.79%
- Accuracy on using their technique (no attack): 69.23%,

- After 5 iterations of BFA, the accuracy of baseline model degrades to below 1%
- Their technique maintains the accuracy (after attack) to 60%